

Section- Generalized Linear Models

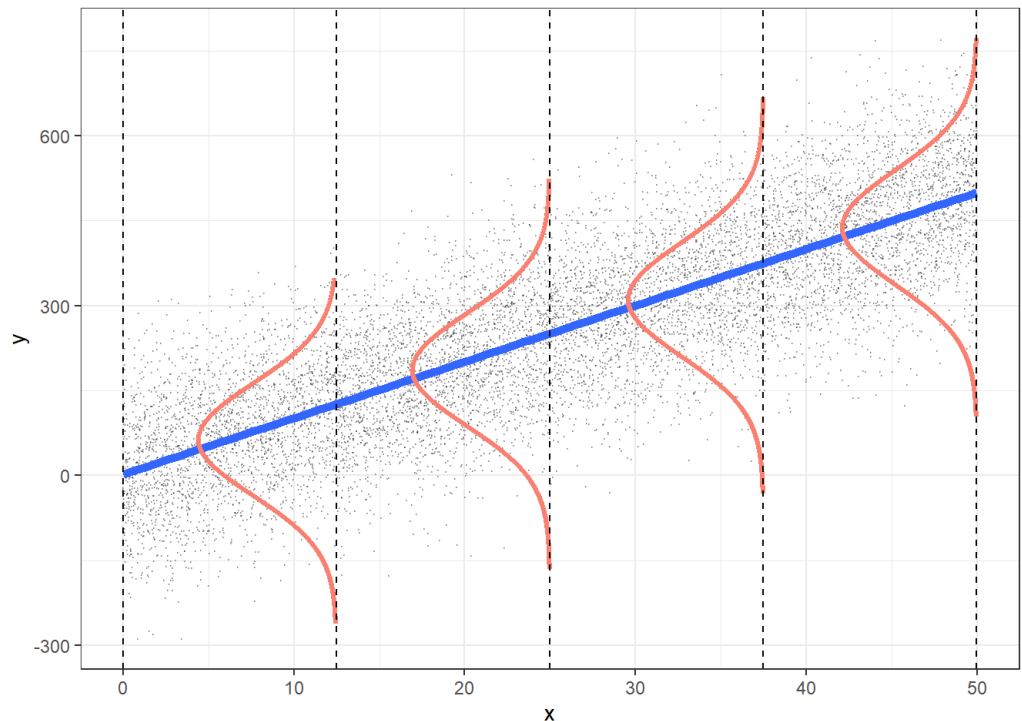
--Case study for Poisson Regression and beyond

Review: Assumptions for Linear Least Squares Regression (OLS)

1. There is a **linear** relationship between the mean response (Y) and the explanatory variable ($\vec{X}$),
2. The errors are **independent**—there's no connection between how far any two points lie from the regression line,
3. The responses are **normally** distributed at each level of $\vec{X}$, and
4. The **variance** or, equivalently, the standard deviation of the responses is equal for all levels of $\vec{X}$.

Equivalently, in formula:

1. $Y = \vec{\theta}^T \vec{X} + \epsilon$
2. ϵ_i independent
3. $Y|X \sim \text{Normal}(\vec{\theta}^T X, \sigma^2)$
or $\epsilon \sim \text{Normal}(0, \sigma^2)$
4. $\sigma^{(i)} = \sigma$



Possible Examples of OLS:

- Reaction times y and car radios.
- Crop yield y and rainfall.
- Heights of sons y and fathers

Possible Examples of Not OLS:

(P/F) Grades y and studying. (Binary)

Family size y and income (Poisson)

Weight y , and exercise (standard deviation may differ)

Surgery outcome (scale 1-10) and patient age (independence)

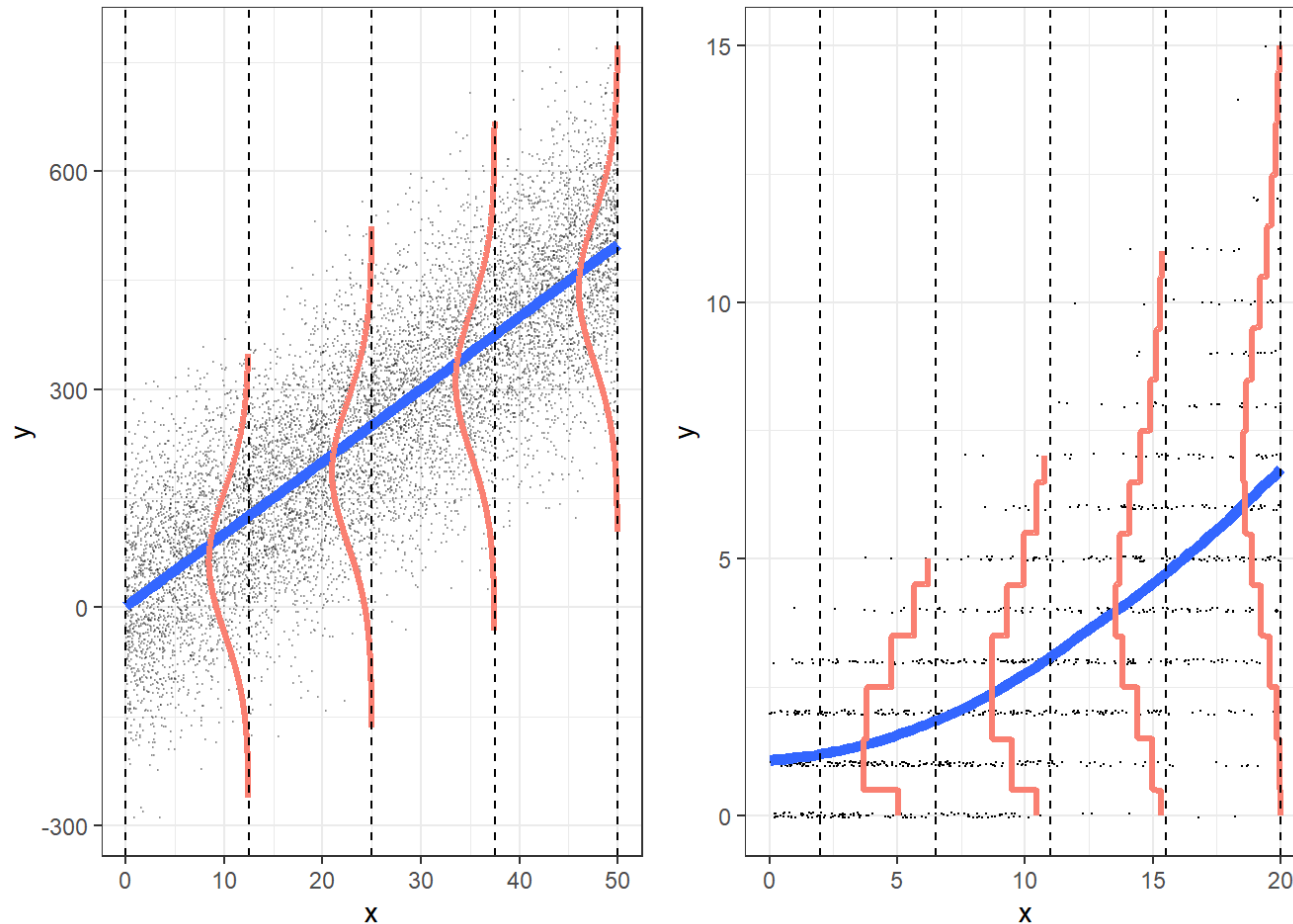
❖ Regression models for counts

Poisson regression is a popular method used in predict variable consists of “count data.”

Poisson Regression Assumptions:

- 1. Poisson Response** The response variable y is a count per unit of time or space, described by a Poisson distribution.
- 2. Independence** The observations must be independent of one another.
- 3. Mean=Variance** By definition, the mean of a Poisson random variable must be equal to its variance.
- 4. Linearity** The log of the mean rate, $\log(\lambda)$, must be a linear function of $\vec{x}$.

Ordinary Linear regression v.s. Poisson regression



For Poisson regression, 1. small values of λ are associated with a distribution that is noticeably skewed with lots of small values and only a few larger ones. 2. variance=mean, 3. logs of the means follow a line.

Examples: using Poisson Regression

1. Are the number of motorcycle deaths in a given year related to a state's helmet laws?
2. Does the number of employers conducting on-campus interviews during a year differ for public and private colleges?
3. Does the daily number of asthma-related visits to an Emergency Room differ depending on air pollution indices?
4. Has the number of deformed fish in randomly selected Minnesota lakes been affected by changes in trace minerals in the water over the last decade?
5. Goals in a soccer match
6. Number of earthquakes in a year
7. The number of accidents during rush hour in the morning
8. The number of calls to a call center during business hours

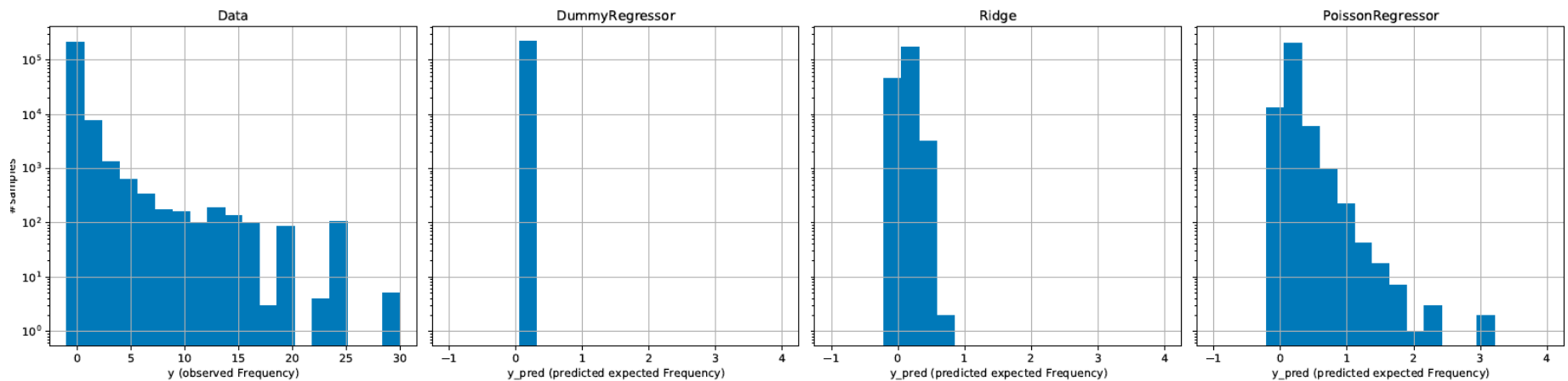
Example: Predicting insurance claims

Goal: predict the **expected number of insurance claims per year** following car accidents.

The dataset ($\mathcal{D} = \{(\vec{x}^{(i)}, y^{(i)}), i = 1, \dots, n\}$) consists of $n = 678,000$ examples with **9 features**, such as driver age, vehicle age, vehicle power, etc. So, $\vec{x}^{(i)} \in \mathbb{R}^9$

The target y is the **frequency** of claims, which is the **number of claims per policy divided by the exposure** (i.e., the duration of the policy in years).

Predictions of insurance claim rates on the test set.



(a) Data. (b) Constant predictor. (c) Linear regression. (d) Poisson regression.

We see that for 94% of the policies, no claims are made, so the data has lots of 0s, as is typical for count and rate data. The average frequency of claims is 10% as in Figure(a).

This can be converted into a dummy model, which always predicts this constant. This results in the predictions shown in Figure (b). The goal is to do better than this.

A simple approach is to use linear regression, combined with some simple feature engineering (binning the continuous values, and one-hot encoding the categoricals). (We use a small amount of l_2 regularization, so technically this is ridge regression.) This gives the results shown in Figure (c). This is better than the baseline, but still not very good. In particular, it can predict negative outcomes, and fails to capture the long tail.

We can do better using Poisson regression, using the same features but a log link function. The results are shown in Figure (d). We see that predictions are much better.

Q: How to quantify performance in this kind of problem?

Performance metrics on the test set.

MSE = mean squared error.

MAE = mean absolute error.

Deviance = Poisson deviance.

Name	MSE	MAE	Deviance
Deviance	0.564	0.189	0.625
Ridge	0.560	0.177	0.601
Poisson	0.560	0.186	0.594

If we use mean squared error, or mean absolute error, we may conclude that ridge regression is better than Poisson regression, but this is clearly not true, as shown in Figure. Instead it is more common to measure performance using the **deviance**, which is defined next.

Deviance is defined as

$$D(y, \hat{\mu}) := 2 \sum_i \left(\log p(y^{(i)} | \mu_*^{(i)}) - \log p(y^{(i)} | \mu^{(i)}) \right)$$

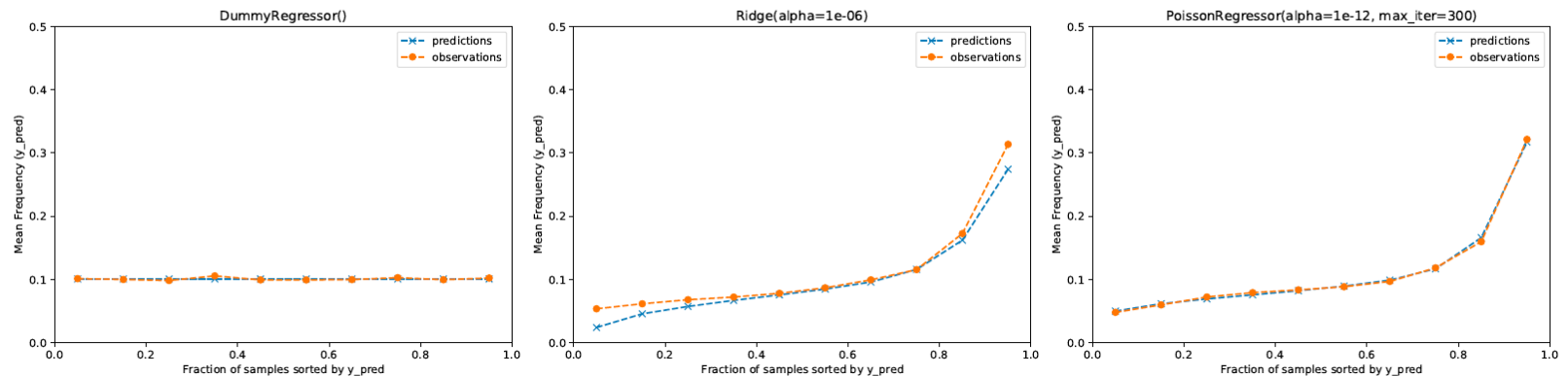
where $\mu^{(i)}$ is the predicted parameters for the i 'th example (based on the input features $x^{(i)}$ and the training set D), and $\mu_*^{(i)}$ is the optimal parameter estimated by fitting the model just to the true output $y^{(i)}$

In the case of Poisson regression, $\mu_^{(i)} = y^{(i)}$*

$$\begin{aligned} D(y, \hat{\mu}) &= 2 \sum_i \left((y^{(i)} \log y^{(i)} - y^{(i)} - \log(y^{(i)}!)) - (y^{(i)} \log \hat{\mu}_i - \hat{\mu}_i - \log(y^{(i)}!)) \right) \\ &= 2 \sum_i \left(y^{(i)} \log \frac{y^{(i)}}{\hat{\mu}_i} - \hat{\mu}_i - y^{(i)} \right) \end{aligned}$$

Calibration plot

We can also compute a calibration plot, which plots the actual frequency vs the predicted frequency. To compute this, we bin the predictions into intervals, and then count the empirical frequency of claims for all examples whose predicted frequency falls into that bin.



We see that the constant baseline is well calibrated, but of course it is not very accurate. The ridge model is miscalibrated in the low frequency regime. In particular, it underestimates the total number of claims in the test set to be 10,693, whereas the truth is 11,935.

The Poisson model is better calibrated (i.e., when it predicts examples will have a high claim rate, they do in fact have a high claim rate), and it predicts the total number of claims to be 11,930.

Further Studies: Generalized Linear Models Examples

Here are some real-world examples where generalized linear models can be used to predict continuous response variables based on their probability distribution. The table consists of reference to the SKlearn class which can be used to model the response variables.

Problem Type	Example	Regression Model (Sklearn)
Agriculture / weather modeling	Number of rain events per year	<u>PoissonRegressor</u>
Agriculture / weather modeling	Amount of rainfall per rainfall event	<u>GammaRegressor</u>
Agriculture / weather modeling	Total rainfall per year	<u>TweedieRegressor</u>
Risk modeling / insurance policy pricing	No of claim events / policyholder per year	PoissonRegressor
Risk modeling / insurance policy pricing	Cost per claim event	GammaRegressor
Risk modeling / insurance policy pricing	Total cost per policyholder	TweedieRegressor
Predictive maintenance	Number of production interruption events per year	PoissonRegressor
Predictive maintenance	Duration of interruption	GammaRegressor
Predictive maintenance	Total interruption time per year	TweedieRegressor

Other regression models for counts:

- Negative Binomial regression model. (mean not equal to variance)
- The Generalized Poisson Regression Model (mean not equal to variance)
- Ordered Logit,
- Ordered Probit
- Nonlinear Least Squares models.

More machine learning techniques such as:

- Complex variants of the Poisson regression model, e.g., the Zero-inflated model.
- The hurdle model
- A Random Forest based regression model
- A Long-Short Term Memory (LSTM) neural network based regression model

References:

1. Cameron A. C. and Trivedi P. K., Regression Analysis of Count Data, Second Edition, Econometric Society Monograph No. 53, Cambridge University Press, Cambridge, May 2013.

2. **[Murphy 1] 12.5**

3. **Poisson regression and non-normal loss (SK-Learn)**

https://scikit-learn.org/stable/auto_examples/linear_model/plot_poisson_regression_non_normal_loss.html

4. **StatsModels**

https://www.statsmodels.org/dev/examples/notebooks/generated/generic_mle.html

5. **Matlab:**

<https://www.mathworks.com/help/stats/fitting-data-with-generalized-linear-models.html>

6. **R**

<https://bookdown.org/roback/bookdown-BeyondMLR/ch-poissonreg.html>